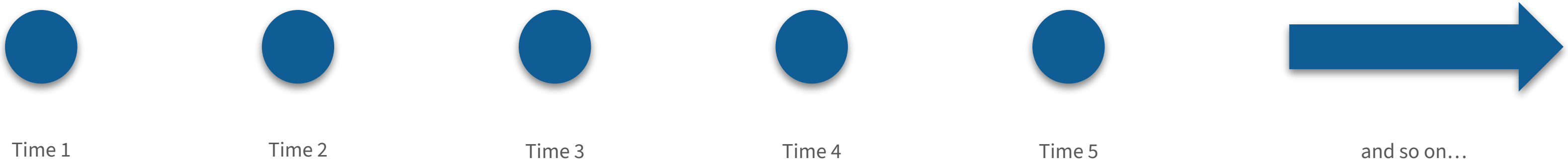
GeeksforGeeks

What is a Sequence?

Transformers – Conceptual Foundations

In the context of data science, sequences are any data that observe a chronological order.



Sequence are everywhere!

- Time series data, like stock market prices. 
- Speech, which is a sequence of phenomes. 
- Music, a sequence of notes. 
- Text, a sequence of words. 
- Video, which can be seen as a sequence of image frames. 

Individual units interact to create meaning.

The dog jumped over the boy, and it fell on the grass.

Individual units interact to create meaning.

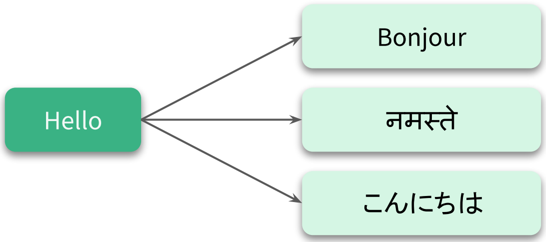
The dog jumped over the boy, and it fell on the grass.

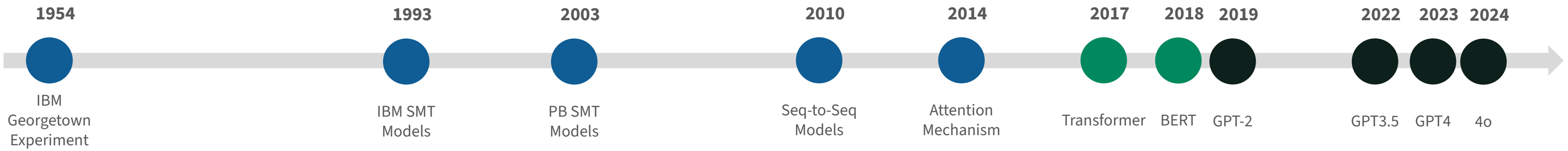
GeeksforGeeks

History of Machine Translation

Transformers – Conceptual Foundations

Convert one language to another



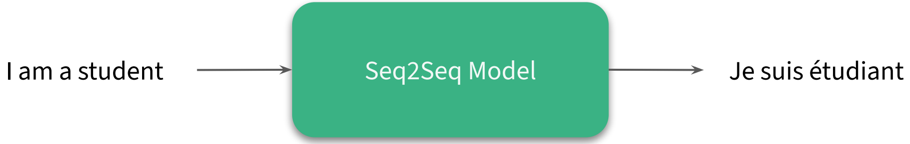


GeeksforGeeks

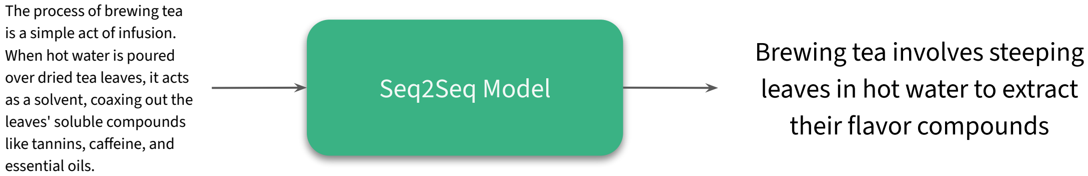
What are Seq2Seq Models?

Transformers – Conceptual Foundations

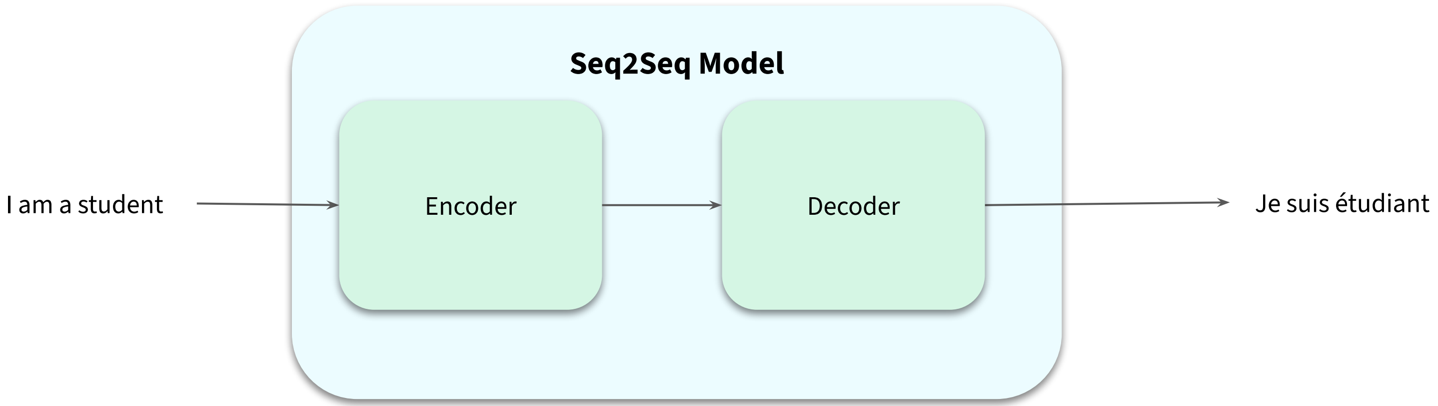
Machine Translation



Summarization



Chatbots

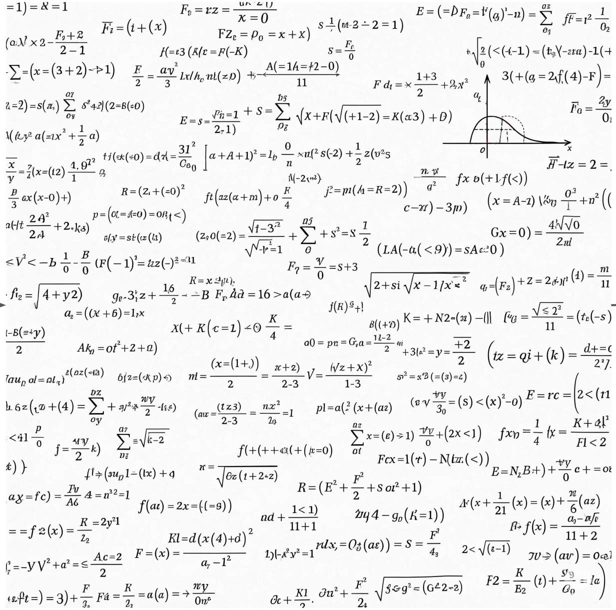


Seq2Seq Model
I am a student
Encoder
Decoder
Je suis étudiant

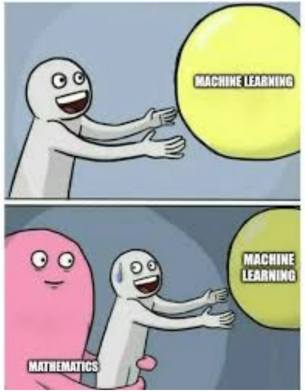
Tangent

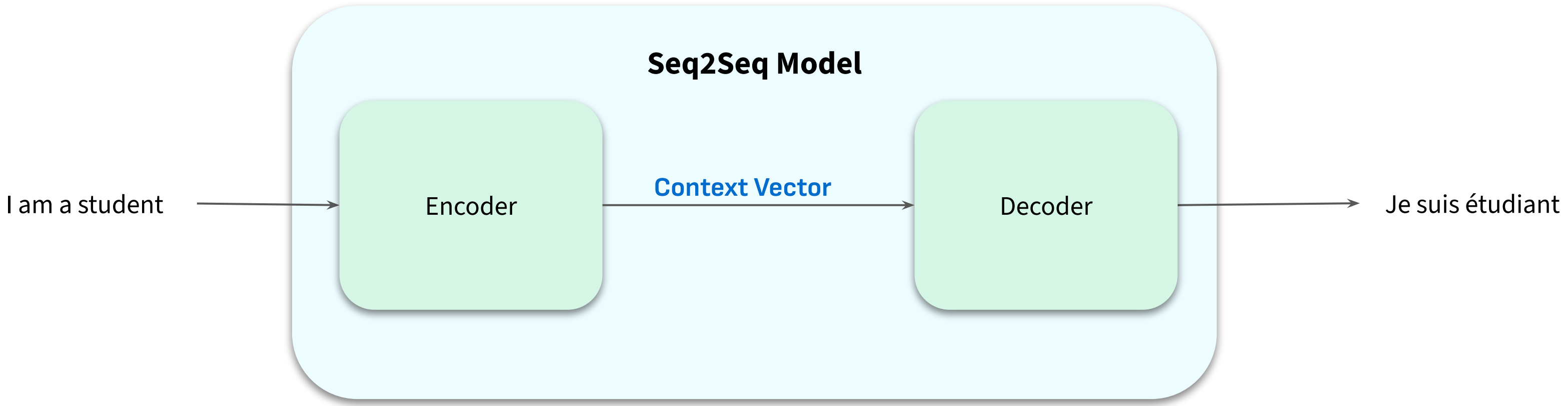
All AI models are mathematical equations under the hood!

I am a student



Je suis étudiant

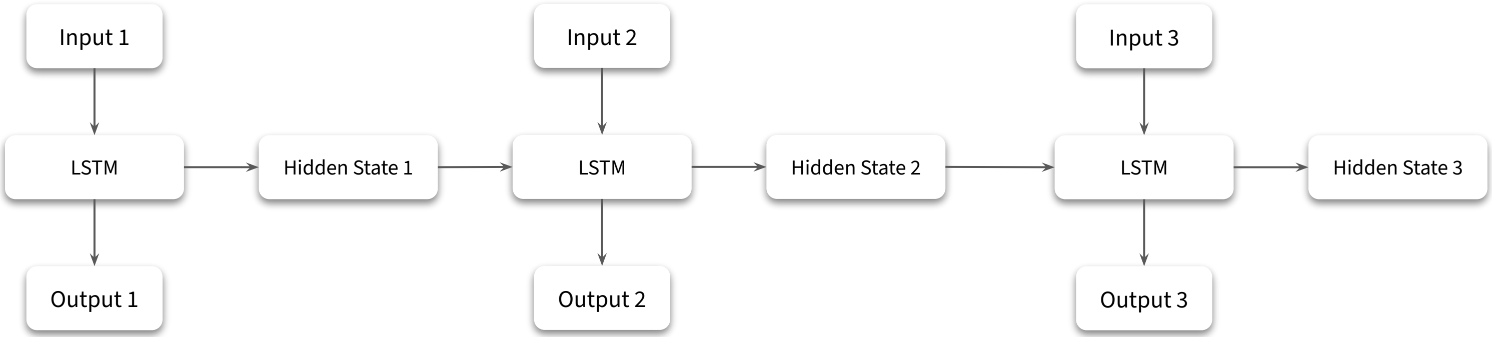



Seq2Seq Model
I am a student
Encoder
Context Vector
Decoder
Je suis étudiant

What are RNN-based Seq2Seq Models?

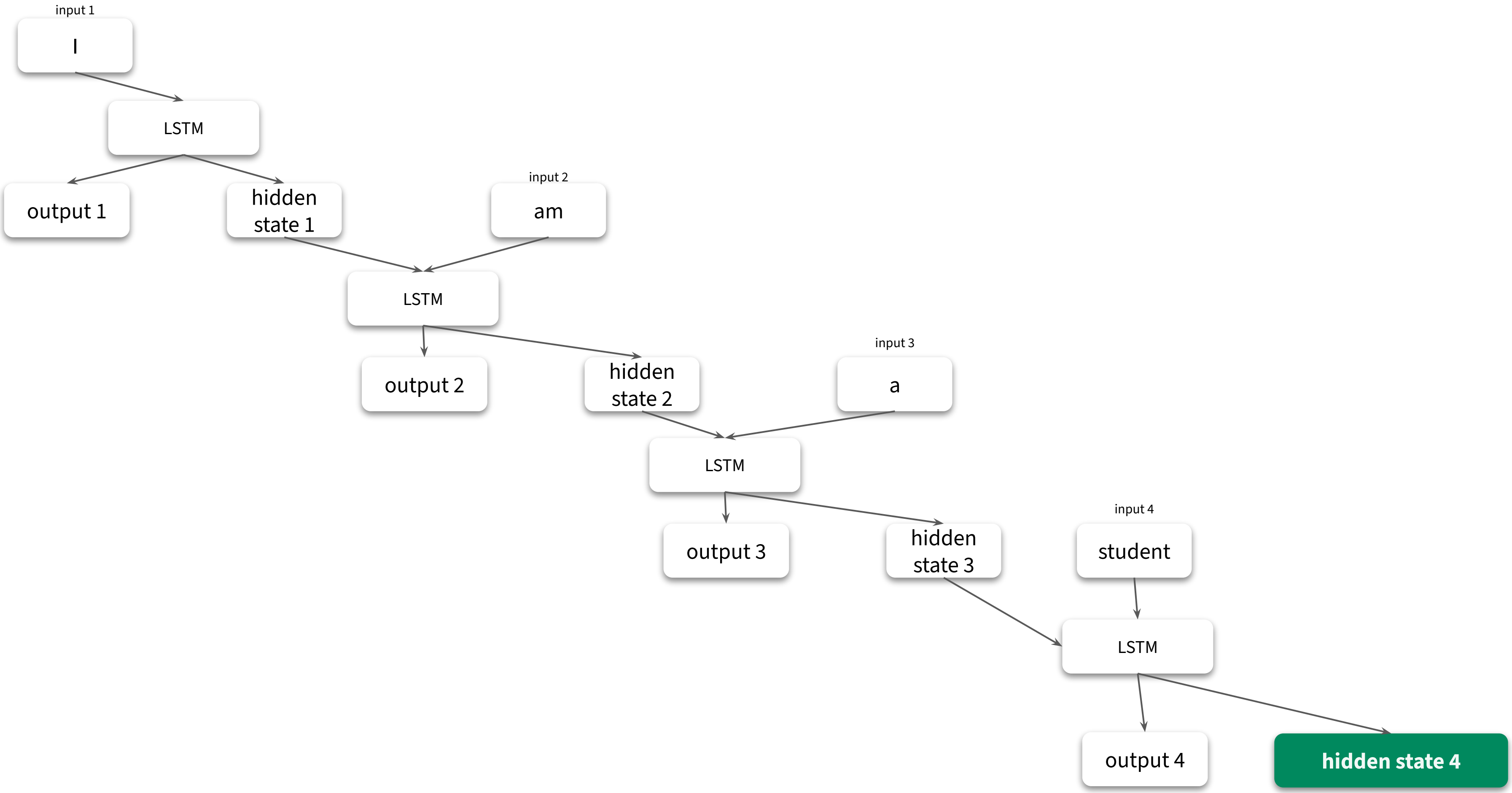
Transformers - Conceptual Foundations

Recurrent Neural Networks



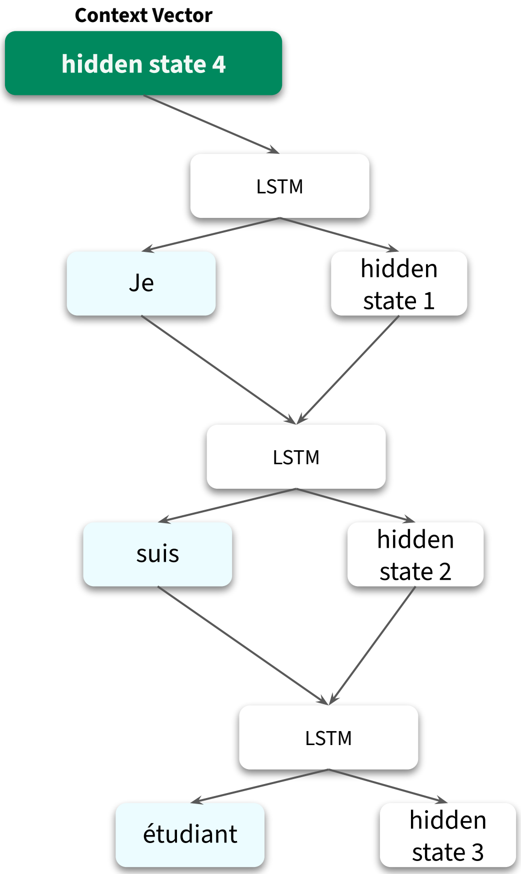
and so on..

Encoder RNN



input 1
I
LSTM
output 1
hidden state 1
input 2
am
LSTM
output 2
hidden state 2
input 3
a
LSTM
output 3
hidden state 3
input 4
student
LSTM
output 4
hidden state 4

Decoder RNN

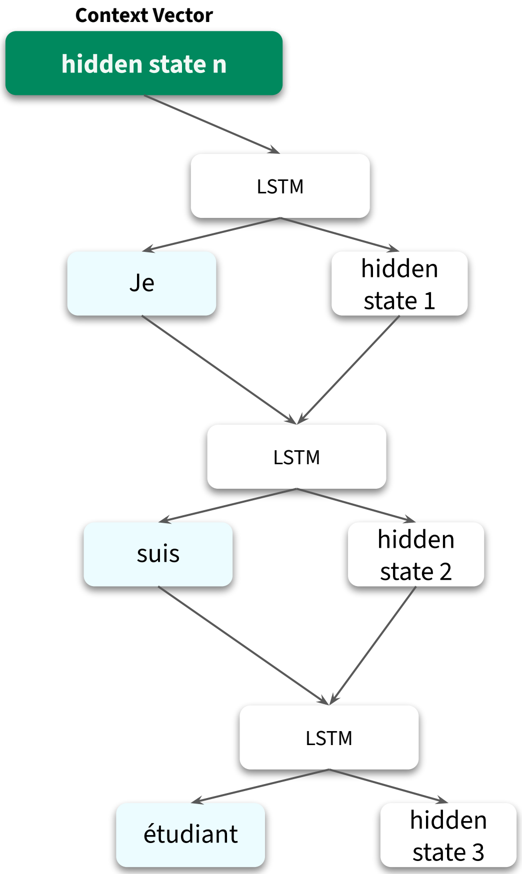

Context Vector
hidden state 4
LSTM
Je
hidden state 1
LSTM
suis
hidden state 2
LSTM
étudiant
hidden state 3

Will this perform well with long sequences? No.

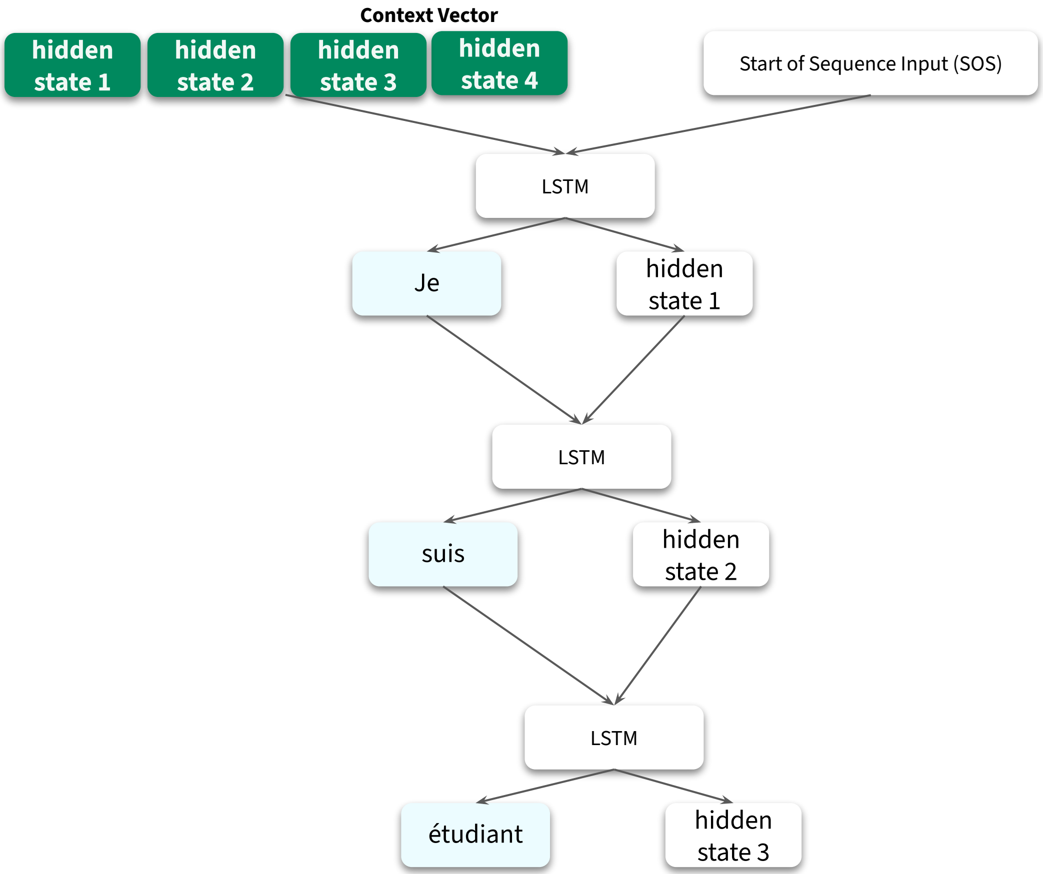
What is Attention?

Transformers – Conceptual Foundations

Decoder RNN without Attention



Decoder RNN with Attention



Interpretability with Attention

	I	am	a	student
Je	**Encoder hidden state 1**	Encoder hidden state 2	Encoder hidden state 3	Encoder hidden state 4
suis	Encoder hidden state 1	**Encoder hidden state 2**	**Encoder hidden state 3**	Encoder hidden state 4
étudiant	Encoder hidden state 1	Encoder hidden state 2	**Encoder hidden state 3**	**Encoder hidden state 4**

Attention != Transformer

Attention ~~Model~~ Mechanism.

Your First High-Level Look at Transformers

Transformers – Conceptual Foundations

Attention is all you need!

Transformers uses RNNs?

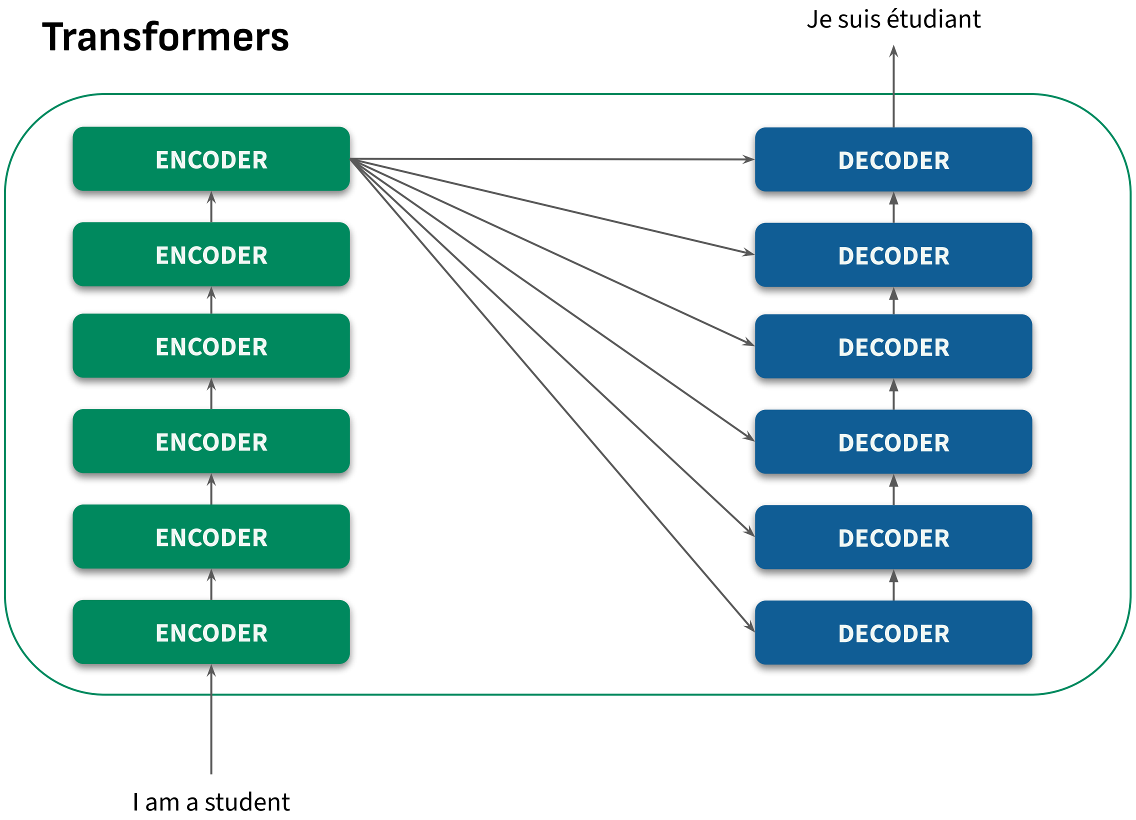
Attention is all you need!

Transformers uses ~~RNNs~~ Attention?

Attention is all you need!

Transformers uses ~~RNNs~~ ~~Attention~~ Self-Attention

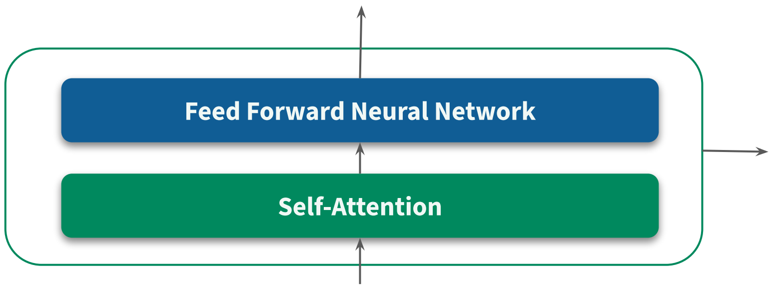
Transformers



The Encoding Component

Transformers – Conceptual Foundations

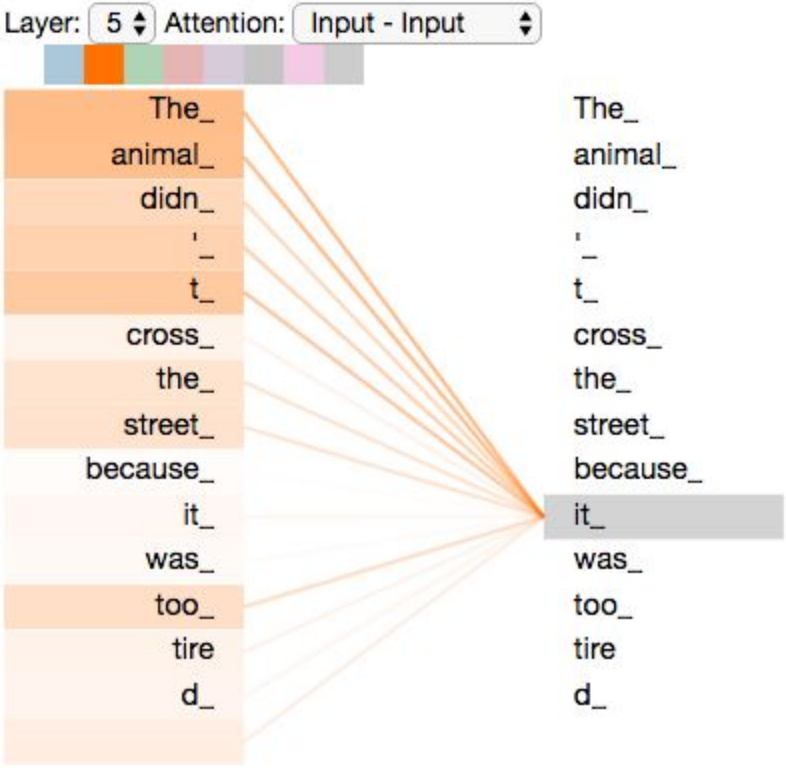
Encoders



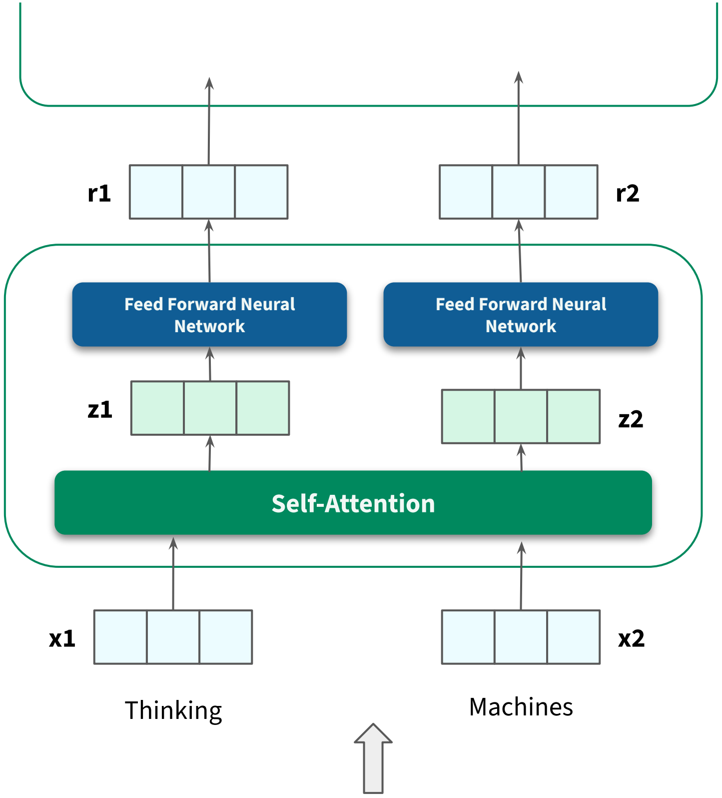
Self-Attention

Transformers – Conceptual Foundations

Self-Attention



Encoders



Thinking machines

Questions

Why is the data going to one single self-attention layer rather than two separate self-attention layers?

Why have the feed forward network layers, at all?

How are attention scores calculated?

How Attention Scores are Calculated?

Transformers – Conceptual Foundations

The Red Fox

The red fox, a creature of cunning and adaptability, is a common sight in both rural and urban environments. Despite its name, this mammal can be found in a variety of colors, including

Value silver, black, and other variations.

Key

Besides red, what is one other colour a red fox can be?

Query

The Red Fox

The red fox, a creature of cunning and adaptability, is a common sight in both rural and urban environments. Despite its name, this mammal can be found in a variety of colors, including silver, black, and other variations. The most iconic morph, however, presents a fiery red coat that contrasts sharply with its white underbelly, black-tipped ears, and dark legs. Its long, bushy tail, often tipped with a distinctive white blaze, serves multiple purposes: as a warm covering in cold weather, a signal flag to other foxes, and a counterbalance for agile maneuvering.

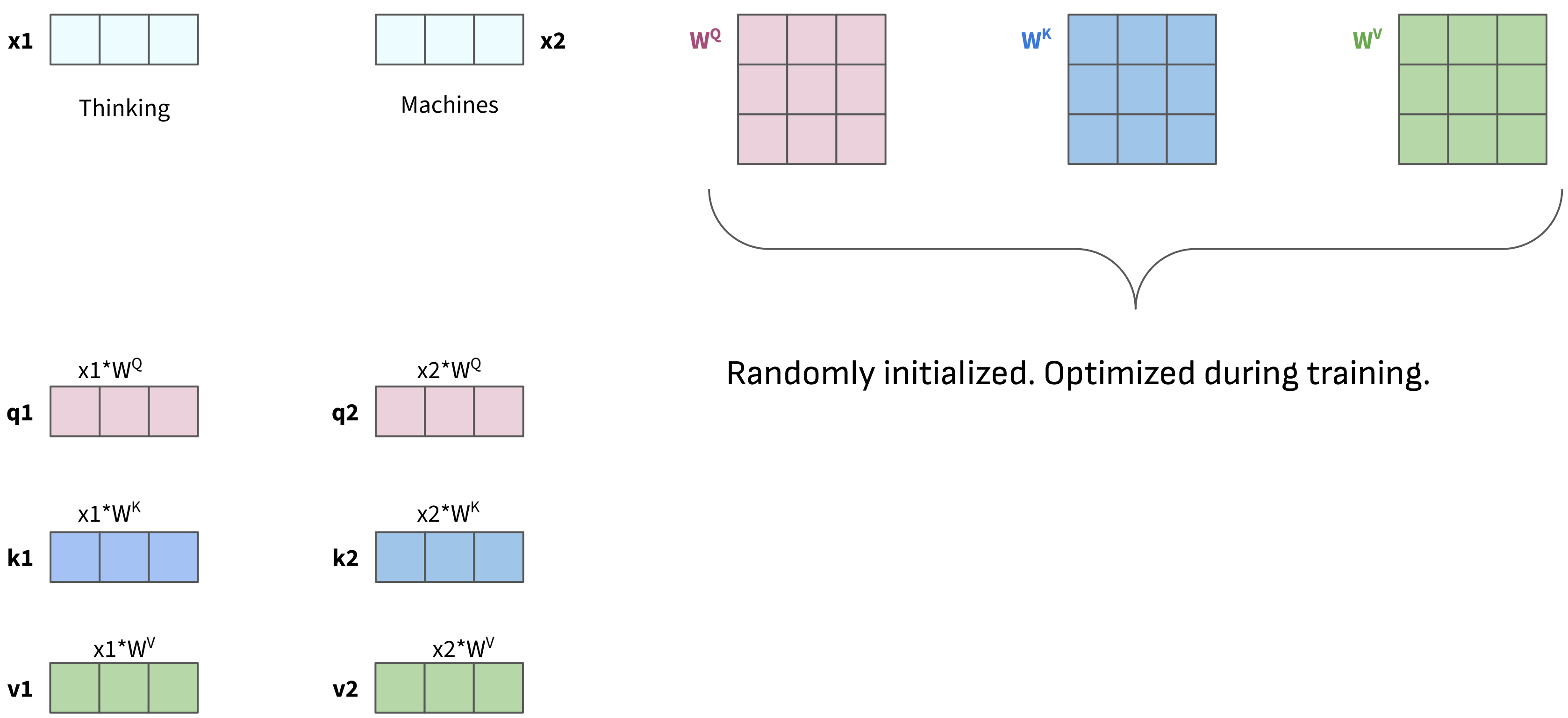
Scientifically known as *Vulpes vulpes*, the red fox boasts the widest geographical distribution of any member of the order Carnivora. Its range extends across the entire Northern Hemisphere, from the Arctic Circle to North Africa, North America, and Eurasia. This remarkable success is a testament to its incredible ability to thrive in a vast array of habitats, from dense forests and open grasslands to harsh deserts and even the heart of bustling cities. In recent decades, red foxes have become increasingly common in suburban and urban areas, where they have proven adept at navigating human-dominated landscapes and exploiting new food sources.

The diet of the red fox is as varied as its habitat. An opportunistic omnivore, it preys primarily on small rodents like mice and voles, as well as rabbits, squirrels, and birds. However, it will readily supplement its diet with fruits, berries, insects, and carrion. In urban settings, foxes are notorious for scavenging from garbage cans and compost heaps, a behavior that highlights their resourcefulness. Their hunting techniques are a marvel of instinct and intelligence. With keen hearing, they can locate prey moving underground and will often perform a characteristic high pounce, leaping into the air to surprise and pin down their meal.

Socially, red foxes are generally solitary hunters, though they live in small family groups in underground dens, often referred to as "earths." These dens can be complex, with multiple entrances and chambers, and are sometimes inherited and expanded over generations. A dominant pair, the dog fox and the vixen, will mate for life and raise a litter of kits each spring. The kits, born blind and deaf, are cared for by both parents and sometimes by older siblings from a previous litter who act as "helpers." By autumn, the young foxes will have dispersed to establish their own territories.

The relationship between humans and red foxes is complex and multifaceted. In many cultures, the fox is a symbol of cunning and trickery, a prominent figure in folklore and fables. While they can be a nuisance to farmers due to occasional predation on poultry, they also play a crucial role in controlling populations of agricultural pests. Their growing presence in cities has led to a mixture of delight and concern, with many people enjoying the sight of these wild creatures while others worry about potential conflicts. Regardless of human perception, the red fox continues to demonstrate a remarkable resilience, a testament to its evolutionary prowess and its ability to adapt to a changing world.

Attention Scores: Setup



Attention Scores: Softmax Scores

Input	Thinking	Machines
Embeddings	x1	x2
Query	$x1*W^Q$ q1	$x2*W^Q$ q2
Key	$x1*W^K$ k1	$x2*W^K$ k2
Value	$x1*W^V$ v1	$x2*W^V$ v2
Score	**q1 . k1 = 112**	**q1 . k2 = 96**
Divide by 8 (scaling factor)	**112/8 = 14**	**96/8 =12**
Softmax Score	**0.88**	**0.12**

$$z1 = \underbrace{0.88 * v1}_{\boxed{s1}} + \underbrace{0.12 * v2}_{\boxed{s2}}$$

Attention Scores

Input	Thinking	Machines
Embeddings	**x1**	**x2**
Query	$x1*W^Q$ **q1**	$x2*W^Q$ **q2**
Key	$x1*W^K$ **k1**	$x2*W^K$ **k2**
Value	$x1*W^V$ **v1**	$x2*W^V$ **v2**
Score	**q1 . k1 = 112**	**q1 . k2 = 96**
Divide by 8	**112/8 = 14**	**96/8 =12**
Softmax Score	**0.88**	**0.12**
Softmax Score * Value	**s1**	**s2**
Attention Score	**z1**	

Why calculate attention score now?

Contextual Grounding Before Processing

Setting up the Feed-Forward Depth

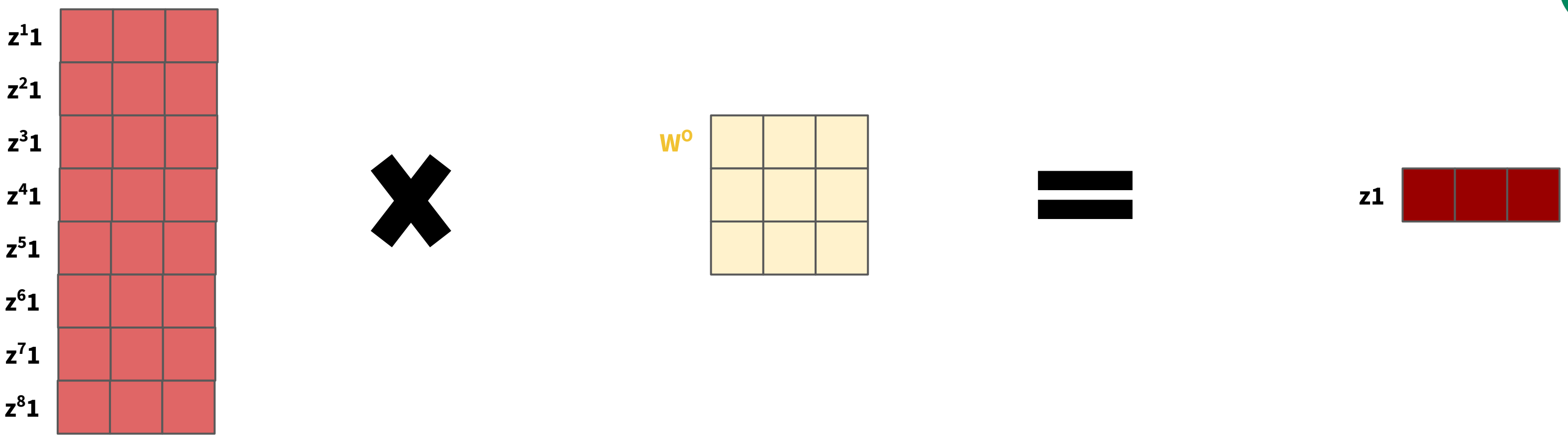
GeeksforGeeks

Multi-headed Self-Attention

Transformers - Conceptual Foundations

Each Head = individual, independent self-attention execution.

Each Head = different way or “lens” to view the same input sequence.



z11
z21
z31
z41
z51
z61
z71
z81
WO
z1

Tangent

Multi-Headed Attention = Multi-Headed Self-Attention

GeeksforGeeks

Positional Encoding

Transformers – Conceptual Foundations

Final Encoding = Embedding + Position Encoding

Input	Thinking	Machines
Embeddings	**x1**	**x2**
Position	**1**	**2**
Positional Encoding	**t1**	**t2**
Final Encoding (Embedding with Time Signal)	x1+t1 **x1**	x2+t2 **x2**

For each position pos and dimension i:

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right)$$

$$PE(pos, 2i+1) = \cos\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right)$$

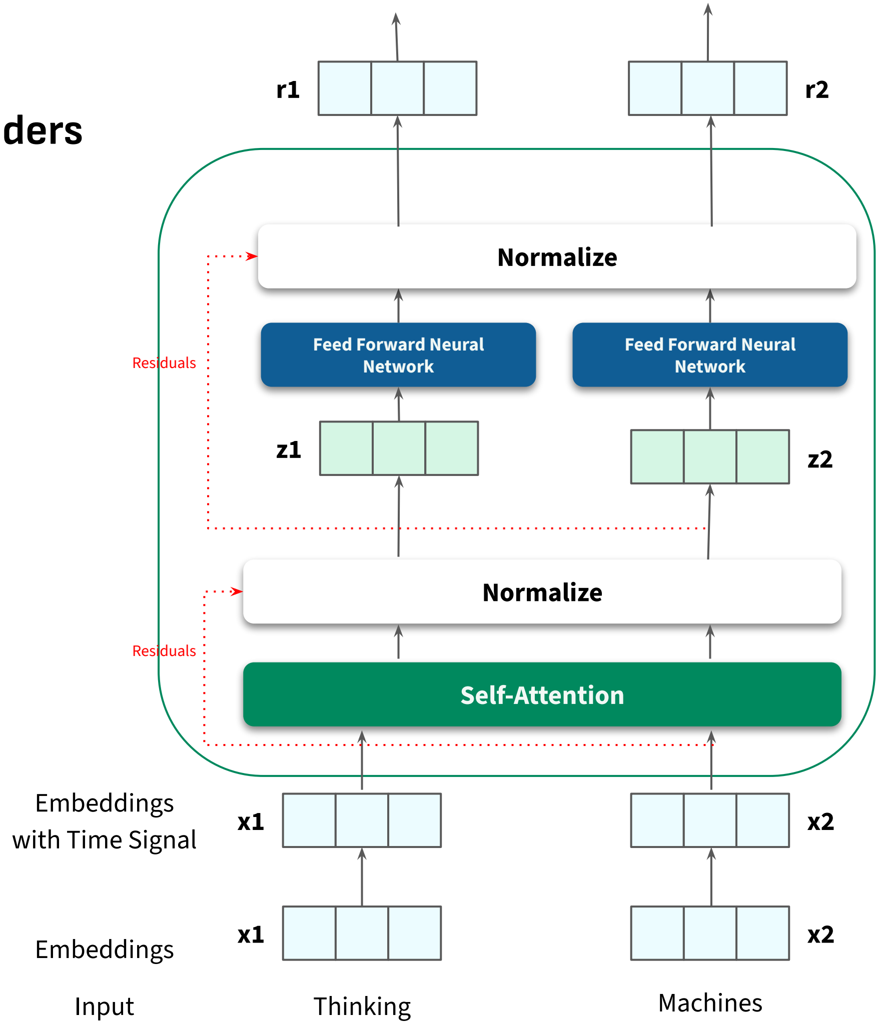
Here, d_{model} is the dimension of the model.

GeeksforGeeks

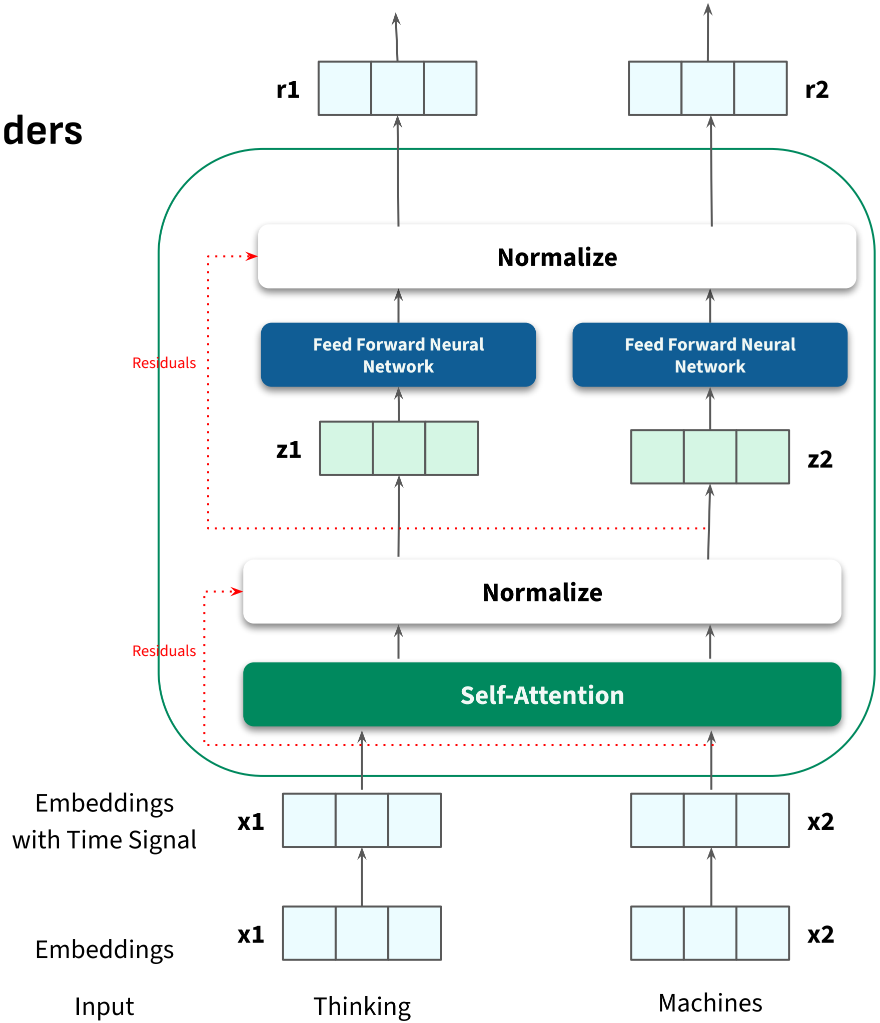
Residuals & Layer Normalization

Transformers – Conceptual Foundations

Encoders


r1
r2
Normalize
Residuals
Feed Forward Neural Network
Feed Forward Neural Network
z1
z2
Normalize
Residuals
Self-Attention
Embeddings with Time Signal
x1
x2
Embeddings
x1
x2
Input
Thinking
Machines

Encoders

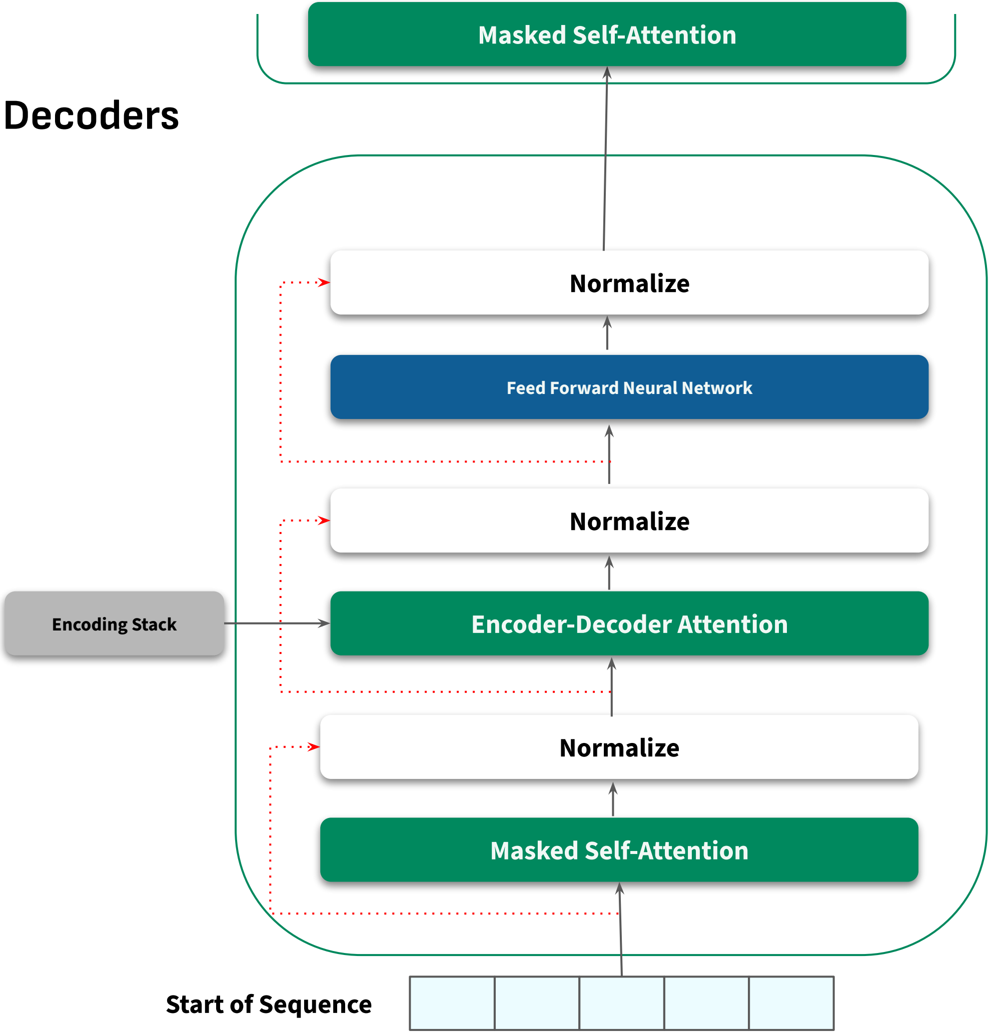

r1
r2
Normalize
Residuals
Feed Forward Neural Network
Feed Forward Neural Network
z1
z2
Normalize
Residuals
Self-Attention
Embeddings with Time Signal
x1
x2
Embeddings
x1
x2
Input
Thinking
Machines

GeeksforGeeks

Decoders

Transformers – Conceptual Foundations

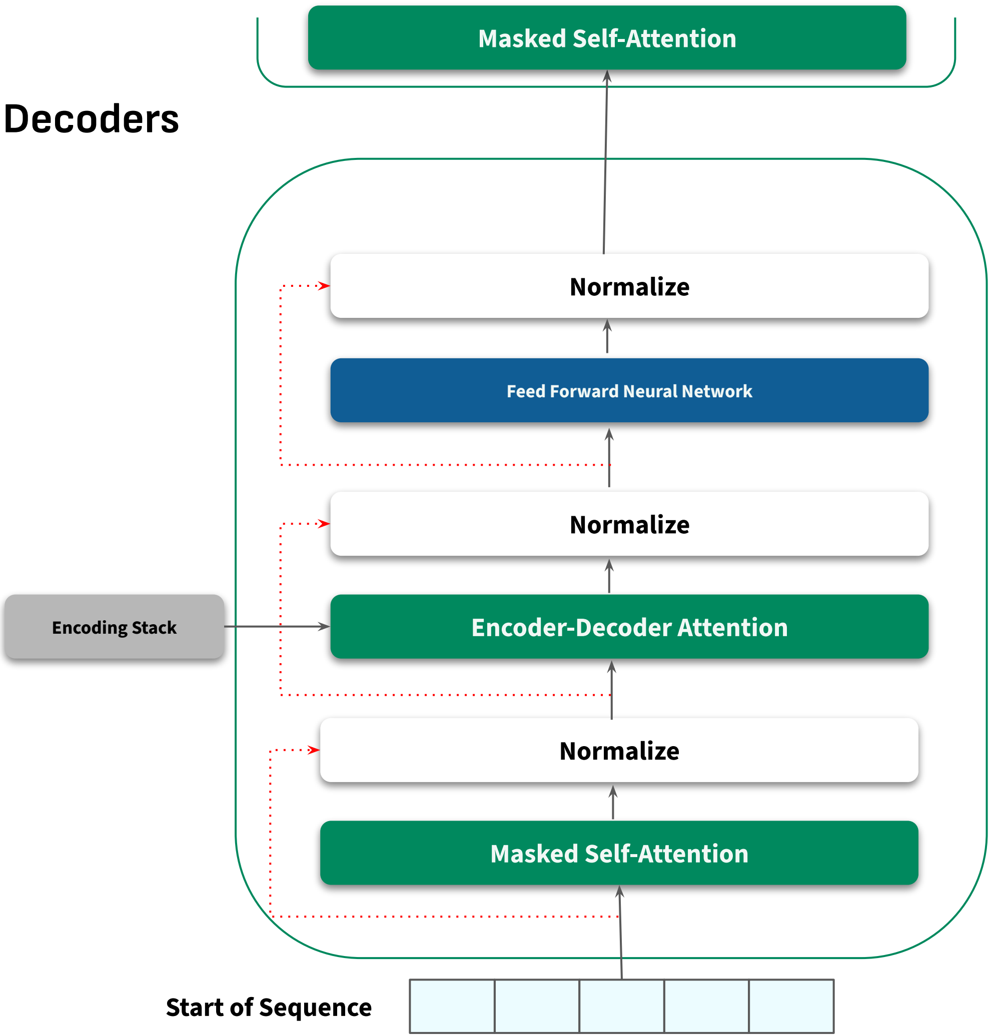
Decoders



The Masked Self-Attention layer allows the decoder to weigh the importance of the different words in the output sequence generated so far.

Which part of the input is most important for me to look at right now to generate the next token?

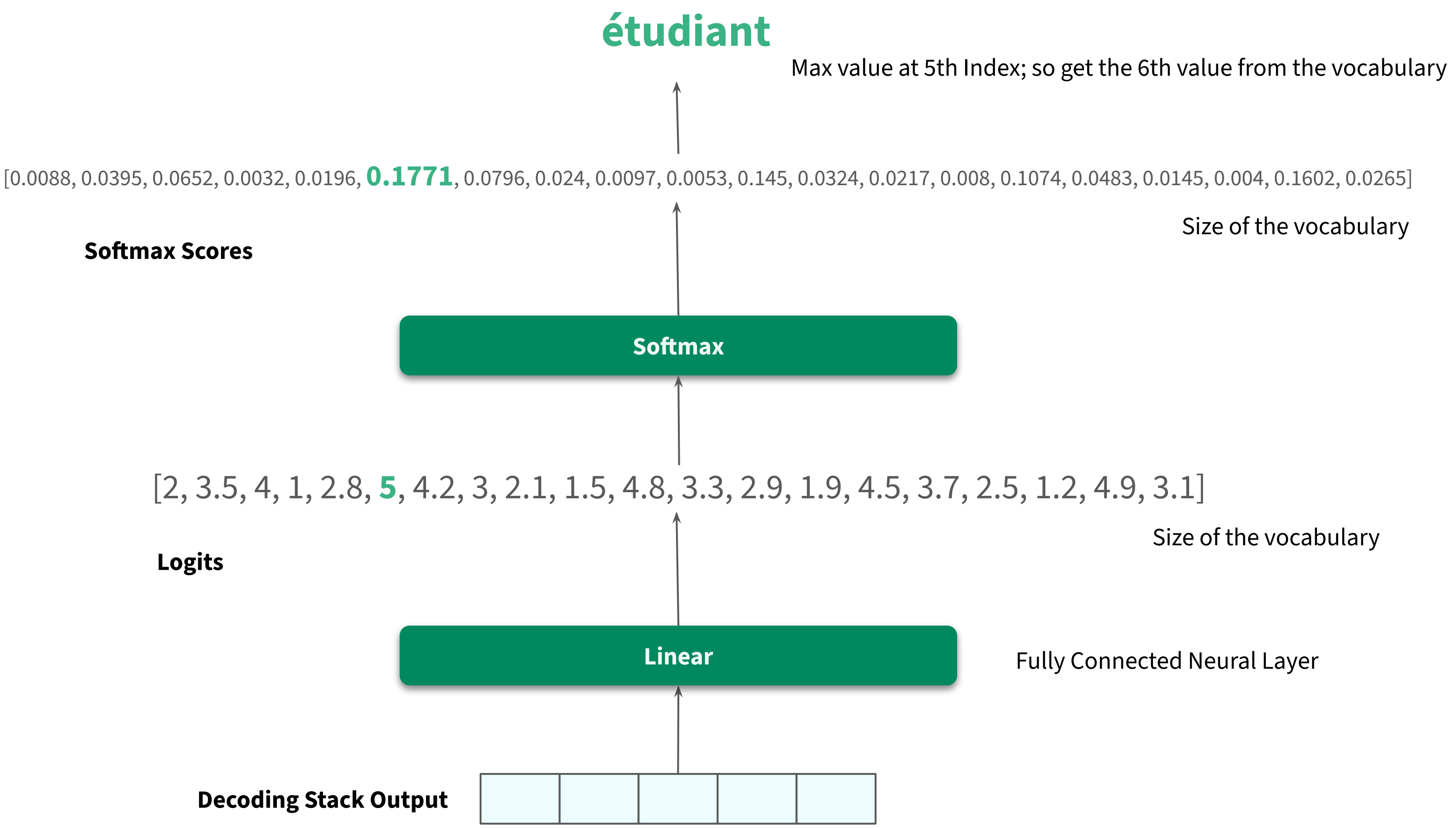
Decoders



Masked Self-Attention
Normalize
Feed Forward Neural Network
Normalize
Encoding Stack
Encoder-Decoder Attention
Normalize
Masked Self-Attention
Start of Sequence

Linear and Softmax Layers

Transformers – Conceptual Foundations

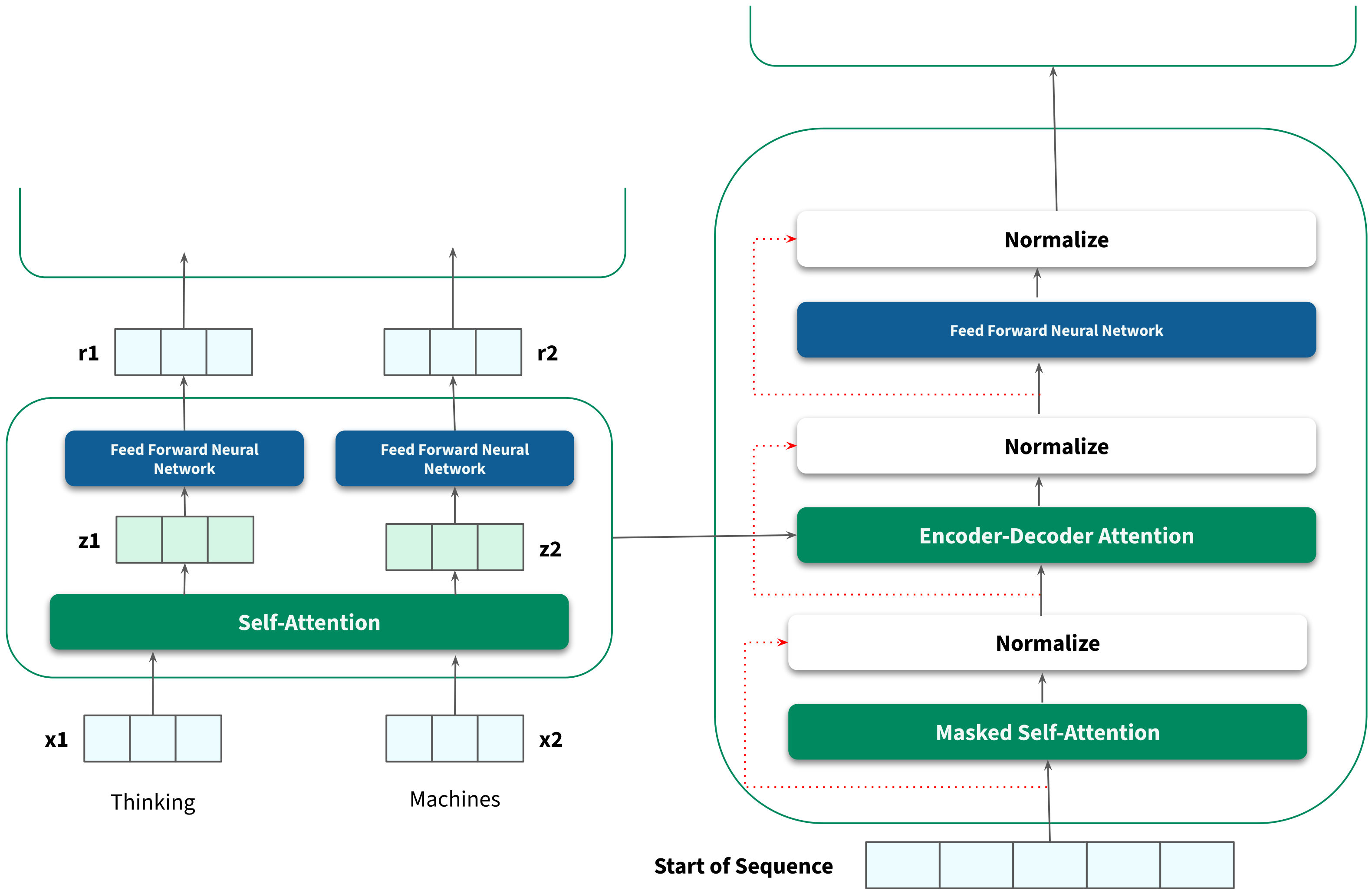


étudiant
Max value at 5th Index; so get the 6th value from the vocabulary
[0.0088, 0.0395, 0.0652, 0.0032, 0.0196, 0.1771, 0.0796, 0.024, 0.0097, 0.0053, 0.145, 0.0324, 0.0217, 0.008, 0.1074, 0.0483, 0.0145, 0.004, 0.1602, 0.0265]
Size of the vocabulary
Softmax Scores
Softmax
[2, 3.5, 4, 1, 2.8, 5, 4.2, 3, 2.1, 1.5, 4.8, 3.3, 2.9, 1.9, 4.5, 3.7, 2.5, 1.2, 4.9, 3.1]
Size of the vocabulary
Logits
Linear
Fully Connected Neural Layer
Decoding Stack Output

GeeksforGeeks

Revisiting the High-Level Look

Transformers – Conceptual Foundations


Normalize
Feed Forward Neural Network
Normalize
Encoder-Decoder Attention
Normalize
Masked Self-Attention
Start of Sequence
r1
r2
Feed Forward Neural Network
Feed Forward Neural Network
z1
z2
Self-Attention
x1
x2
Thinking
Machines